batmonitor 0.74: one cycle every 5 s | max persistent decision states 12288 | blue = decision (c/bm_tables.h, make check) | green = matrix / amount | details: charge_flow, discharge_flow, decision_tables

inputs (bm_parse: only valid numbers, else stale)
PCC (live, sofar_fast 4 s), Sofar Bat1 W + SoC, PV, heat pump W, R290 Hz,
BMS SoC / U / CCL / DCL / lowest + highest cell, AC-in / AC-out per phase, grid connected

D7 house on AC-out1 (0.69-c)
HOUSE_ACOUT 0 off / 1 on / 2 auto (> 99 W on, 60 s < 30 W off)
on: winter Z2 near 0 (discharge = AC-out1 load), summer Z1 near 0
Lenggries env: HOUSE_ACOUT=0

D5 PCC > 20 kW for 2 min (at once > 23 kW) -> DO4 pulse
pv_relay/DO4 (FoxESS shedding)
first: R290 boost + air conditioning at ≥ 18 kW
grid offline: AC-out feed > 5 kW / unit for 10 s -> at once

charge path

D4 forecast export cap (May-Aug)
while the forecast expects export > 18 kW:
wait below the cap, charge only above it

S0 caps per phase: chg_cap, dis_cap (charger, BMS CCL / DCL)
top hold (0.73-c, TOP_HOLD_STACK): charge cap so the highest cell stays at TOP_HOLD_CELL_V

block

SoC, caps

D1 Sofar first, S3 source DAY / B1 / STK / WPG
winter heat pump rule (0.61-c): Bat1 discharging -> 0 W, else WP_WINTER_W
make check: 128 cells, 14 outcomes; details wp_matrix

D2 mode per stack, first match:
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > HOLD > PROT > CHARGING > DIS
FORCE = SoC < SOC_FORCE or LOW (DCL 0 / cell < CELL_FORCE_V; ends at SOC_MIN_RELEASE)
CHARGE also at 100 % for force_full (FF) and the top hold window (TOP); HOLD = force_full without surplus
make check: 64512 combinations per stack -> BMS_MISSING 32256 · FORCE 23040 · STALE 4608 · LADESPERRE 1152 · HOLD 1152 · CHARGE 1080 · PROT 432 · CHARGING 360 · DIS 360 · FULL 72

D-lead: stack more than SOC_BALANCE_ON ahead

src, WPMAX

mode matrix BM_MODE_RULE
0 W | FORCE | charge | discharge

CHARGE

FORCE

DIS

charge amount
max(0, own + KP x (PCC + Bat1_eff - target))
target = CHARGE_RESERVE (+ EXPORT_CAP while D4 cap);
≤ sum of chg_cap

FORCE: min(500 W, chg_cap)
any time of day, no TOU window

alloc_levels up to chg_cap
stack by stack in CHARGE_PRIORITY,
force_full stack first, else the lead stack last

0 W rows

discharge branch

BM_SRC_RULE[src]
STK: Bat1 signed, +trickle, idle 0
else: Bat1 charge only, idle 20 W

D3 PROP / IDLE
(D7 on: HOUSE / PROP / EXPCAP instead)

discharge amount
min(PROP ? deficit : idle, dis_max)
dis_max = W_MAX or the winter heat pump cap (WPMAX)
D7 winter: at most what keeps the export at CHARGE_RESERVE (EXPCAP, 0.74-c)

alloc_levels up to dis_cap, all discharging phases
lead stack first (BAL, SPILL / WAIT), else the same share (EQ)
single-unit discharge (D6, 1PH) removed in 0.70-c

setpoints per phase -> Multis (+ charge / - discharge)
rule text: mode SRC: split [CAP] [TOP] [WPMAX(W)] [LOW] [CCL / DCL / CELL]
-> state.json -> victron2db.py (cron 1/min) -> wagodb.pv_victron.bm_l1..3_rule